BERNARDO SUBERCASEAUX

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EDUCATION

Carnegie Mellon University, PA, United States — August 2021 - (expected) 2027
Ph.D. in Computer Science, advised by Marijn J. H. Heule.
Thesis: *SAT Encodings: From Art to Science*.
Committee: Ryan Williams (MIT), Stefan Szeider (TU Wien), Jeremy Avigad (CMU), and Ruben Martins (CMU).

University of Chile, Chile — March 2014 - December 2020
M.Sc. in Computer Science (GPA 7.0/7.0), under the supervision of Pablo Barceló and Jorge Pérez.
Double Bachelor in Engineering and Computer Science (GPA 7.0/7.0), 2019.

CentraleSupélec (ex École Centrale Paris), France — September 2016 - June 2018
Bachelor + M.Eng. in Computer Science, double-degree program.

RESEARCH SUMMARY

My research spans automated reasoning, algorithms, and computational complexity. I use SAT solvers to tackle long-standing open problems in discrete mathematics through finite computation. My central tool is the design of novel SAT encodings: reformulations into propositional satisfiability that can accelerate solving by several orders of magnitude. I have developed theoretical foundations for SAT encodings, studying their size and propagation strength, with a particular focus on constructing efficient encodings automatically. Furthermore, I have made substantial contributions to the theory of explainable AI, and shown how SAT solvers can be used for explaining classic machine learning models.

AWARDS AND HONORS

- **G-Research PhD Prize 2026**, for my thesis proposal.
- **Best Paper Award, LPAR 2026**, for *A SAT Attack on Tarski's High School Algebra Problem*.
- **Distinguished Paper Award, PODS 2025**, for *Explaining k-Nearest Neighbors: Abductive and Counterfactual Explanations*; invited to a special issue of TODS.
- **Best Paper Award, CICM 2025**, for *Automated Symmetric Constructions in Discrete Geometry*.
- **Best Paper Award Runner-Up, CICM 2024**, for *Automated Mathematical Discovery and Verification: Minimizing Pentagons in the Plane*.
- **Best Paper Award, LPAR 2023**, for *Toward Optimal Radio Colorings of Hypercubes via SAT-solving*.
- **Best Paper Award Nomination, TACAS 2023**, for *The Packing Chromatic Number of the Infinite Square Grid is 15*; featured in Quanta Magazine and invited to a special issue of JAR.
- **Spotlight, NeurIPS 2021** ($< 3\%$), for *Foundations of Symbolic Languages for Model Interpretability*.
- **1st place Master's Thesis 2021**, IEEE LA-CCI Latin American Contest of Master's Theses in Artificial Intelligence.
- **Eiffel Excellence Scholarship 2016**, to pursue a double-degree undergrad in France with tuition and a stipend (sole Chilean recipient).

CONFERENCE PUBLICATIONS

30. **LPAR 2026 (Best paper award)**, *A SAT Attack on Tarski's High School Algebra Problem*, with B. Przybocki.
29. **LPAR 2026**, *Coloring Queens with Thousands of Encodings*, with B. Przybocki and M. Heule.

28. **CICM 2026**, *Doubly Saturated Ramsey Graphs: A Case Study in Computer-Assisted Mathematical Discovery*, with B. Przybocki, J. Mackey, and M. Heule.
27. **SAT 2026**, *Automated Reencoding Meets Graph Theory*, with B. Przybocki and M. Heule.
26. **SAT 2026**, *Near-Optimal Encodings of Cardinality Constraints*, with A. Krapivin and B. Przybocki.
25. **STOC 2026**, *Optimal and Efficient Partite Decompositions of Hypergraphs*, with A. Krapivin, B. Przybocki, and N. Sanhueza-Matamala.
24. **FUN 2026**, *A Demigod's Number for the Rubik's Cube*, with A. Merino.
23. **FUN 2026**, *Solving Small Rubik's Cubes as Slowly as Possible*, with J. Quan, N. Kim, and J. Mackey.
22. **FUN 2026**, *Price of Locality in Permutation Mastermind: Are TikTok Influencers Chaotic Enough?* (single author).
21. **PODS 2025 (Distinguished paper award, invited to special issue of TODS)**, *Explaining k-Nearest Neighbors: Abductive and Counterfactual Explanations*, with P. Barceló, A. Kozachinskiy, M. Romero, and J. Verschae.
20. **CICM 2025 (Best paper award)**, *Automated Symmetric Constructions in Discrete Geometry*, with E. Mackey, L. Qian, and M. Heule.
19. **CADE 2025**, *Unfolding Boxes with Local Constraints*, with L. Qian, E. Wang, and M. Heule.
18. **INFOCOM 2025**, *Latency Guarantees for Caching with Delayed Hits*, with K. Gurushankar and N. Singer.
17. **AAAI 2025**, *Probabilistic Explanations for Linear Models*, with M. Arenas and K. Meel.
16. **CG 2024**, *A Mathematical Analysis of PlaceIt: A Game of Perfect Online Sorting*, with P. Ruiz Cuevas and C. Chock.
15. **KR 2024**, *A Uniform Language to Explain Decision Trees*, with M. Arenas, P. Barceló, D. Bustamante, and J. Caraball.
14. **CICM 2024 (Best paper award runner-up)**, *Automated Mathematical Discovery and Verification: Minimizing Pentagons in the Plane*, with J. Mackey, M. Heule, and R. Martins.
13. **ITP 2024**, *Formal Verification of the Empty Hexagon Number*, with W. Nawrocki, J. Gallicchio, C. Codel, M. Carneiro, and M. Heule.
12. **LPAR 2024**, *Sometimes Hoarding is Harder than Cleaning: NP-hardness of Maximum Blocked-Clause Addition* (single author).
11. **FUN 2024**, *PackIt! Gamified Rectangle Packing*, with T. Garrison and M. Heule.
10. **LPAR 2023 (Best paper award)**, *Toward Optimal Radio Colorings of Hypercubes via SAT-solving*, with M. Heule.
9. **TACAS 2023 (Featured in Quanta Magazine, Best paper award nomination, invited to special issue of JAR)**, *The Packing Chromatic Number of the Infinite Square Grid is 15*, with M. Heule.
8. **SAT 2022**, *The Packing Chromatic Number of the Infinite Square Grid is at Least 14*, with M. Heule.
7. **NeurIPS 2022**, *Augmenting Online Algorithms with ε-Accurate Predictions*, with A. Gupta, D. Panigrahi, and K. Sun.
6. **NeurIPS 2022**, *On Computing Probabilistic Explanations for Decision Trees*, with M. Arenas, P. Barceló, and M. Romero.
5. **FUN 2022**, *Wordle is NP-hard*, with D. Lokshtanov.
4. **NeurIPS 2021 (Spotlight, $< 3\%$)**, *Foundations of Symbolic Languages for Model Interpretability*, with M. Arenas, D. Baez, P. Barceló, and J. Pérez.

3. **FUN 2021**, *The Computational Complexity of Evil Hangman*, with J. Barbay.
2. **NeurIPS 2020**, *Model Interpretability through the Lens of Computational Complexity*, with P. Barceló, M. Monet, and J. Pérez.
1. **ICDT 2020**, *On the Expressiveness of LARA: A Unified Language for Linear and Relational Algebra*, with P. Barceló, N. Higuera, and J. Pérez.

JOURNAL PUBLICATIONS

5. **TALG 2026**, *Computing NP-hard Repetitiveness Measures via MAX-SAT*, with H. Bannai, K. Goto, M. Ishihata, S. Kanda, D. Köppl, and T. Nishimoto.
4. **JAIR 2025** (journal version), *On Computing Probabilistic Explanations for Decision Trees*, with M. Arenas, P. Barceló, A. Kozachinskiy, and M. Romero.
3. **NAW 2024**, *A proof long enough to stump Leonhard Euler*, with M. Heule.
2. **TCS 2022** (journal version), *On the expressiveness of Lara: A proposal for unifying linear and relational algebra*, with P. Barceló, N. Higuera, and J. Pérez.
1. **IOI Journal 2016**, *Wavelet Trees for Competitive Programming*, with R. Castro, N. Lehmann, and J. Pérez.

OTHER PUBLICATIONS (WORKSHOPS AND PREPRINTS)

6. **arXiv preprint 2025**, *From the Finite to the Infinite: Sharper Asymptotic Bounds on Norin's Conjecture via SAT*, with M. Kirchweger, T. Peitl, and S. Szeider.
5. **ModRef 2025**, *Asymptotically Smaller Encodings for Graph Problems and Scheduling* (single author).
4. **arXiv preprint 2024**, *Pentagon Minimization without Computation*, with J. Mackey.
3. **arXiv preprint 2024**, *Assortment Optimization For Conference Goodies With Indifferent Attendees*, with F. Gutiérrez.
2. **AFCI@NeurIPS 2020**, *Foundations of Languages for Interpretability and Bias Detection*, with P. Barceló and J. Pérez.
1. **DEEM 2019** & **AMW 2019**, *Expressiveness of Matrix and Tensor Query Languages in terms of ML Operators*, with P. Barceló, N. Higuera, and J. Pérez.

BOOK CHAPTERS

1. ***Interpretability as Query Answering: The Case for Logical Languages***, with M. Arenas and P. Barceló. In *Logic and AI Today: Interfaces After the Deep Learning Revolution*, edited by J. van Benthem, L. Hornischer, T. Icard, and F. Liu. To appear in 2026.

INVITED TALKS

- **LIACS, Leiden University** (online) — August 2026
  Breaking down graphs into structured pieces, optimally and efficiently.
- **MAA MathFest: Computers and Combinatorics**, Boston — August 2026
  SAT Solving for the Discrete Mathematician's Toolbox.
- **Dagstuhl Seminar 26221**, Dagstuhl — May 2026
  Compact CNF Encodings for Boolean Functions.
- **Université Clermont-Auvergne**, Clermont-Ferrand — May 2026
  Breaking down graphs and hypergraphs into structured pieces, optimally and efficiently.
- **Banff International Research Station**, Banff — January 2026
  Towards a Science of CNF Encodings.

- **AGCO Seminar, Pontificia Universidad Católica de Chile**, Santiago (Chile) November 2025
  Optimal d-Clique Decompositions for Hypergraphs.
- **Simons Institute / SLMath**, Berkeley April 2025
  Computer-Assisted Intuition: SAT Solvers in Mathematical Discovery.
- **Joint Mathematics Meetings: 35th Anniversary of AI and Math**, Seattle January 2025
  Automated Reasoning for the Discrete Mathematician.
- **Boston Computation Club**, Boston August 2023
  The Packing Chromatic Number of the Infinite Square Grid.
- **IISER Math Club**, Bhopal July 2023
  The Packing Chromatic Number of the Infinite Square Grid.
- **Stanford University**, Stanford October 2022
  Power to computers, Power to the people: from computer-aided mathematics to formal explainability.
- **Logic-based Methods in Machine Learning (LMML), FLoC**, Haifa July 2022
  The Exciting Theory of Formal Explanations.
- **AGCO Seminar, CMM, Universidad de Chile**, Santiago (Chile) July 2022
  Augmenting Online Algorithms with ε-Accurate Predictions.
- **DataLab UC: Manejo de Datos y Teoría de la Computación** (online) April 2021
  Árboles y diagramas de decisión: operaciones y equivalencias.
- **LINKS Seminar, Inria / Université de Lille** (online) February 2021
  Foundations of Languages for Interpretability.

RESEARCH INTERNSHIPS AND VISITS

- **TU Wien**, Research visit to Prof. Stefan Szeider (Summer 2025)
- **Amazon**, Automated Reasoning Group, supervised by Mike Whalen (Summer 2024)
- **Amazon**, Automated Reasoning Group, supervised by Mike Whalen (Summer 2023)
- **Simons Institute**, Data-Driven Decision Processes (Fall 2022)

TEACHING EXPERIENCE

Teaching assistant for 19 course offerings across 14 distinct courses (CMU, CentraleSupélec, University of Chile, PUC Chile): *Logic and Mechanized Reasoning*, *Undergraduate Quantum Computation*, *Complexity and Computability*, *Design and Analysis of Algorithms*, *Advanced Analysis of Algorithms*, *Complexity Theory*, *Discrete Math*, *Theory of Computation*, *Introduction to Programming*, *Introduction to Algebra*, *Algorithms and Data Structures*, *Databases*, *Programming Languages*, and *Workshop for Competitive Programming.*

Invited lectures:

- *Applications of Satisfiability in Mathematics*, at University of Warwick (Logic and Verification, 2025). Invited by Arnab Bhattacharyya.
- *SAT for Math*, at Carnegie Mellon University (Advanced Topics in Logic: Automated Reasoning and Satisfiability, 2024, 2025, and 2026). Invited by Marijn Heule.

SERVICE

Program Committee for FUN 2026 and AAAI 2025. Helped organize the ModRef workshop 2026. Served as committee member for the master's thesis of Diego Bustamante, at PUC Chile (2024).

Reviewed for TALG, ESA, ITCS, JAIR, SAT, NeurIPS, Communications of the ACM, TACAS, ICLR, and Operations Research. In total, I have reviewed 37 papers.

MENTORING

- Jenny Quan and Noah Kim (2025). Undergraduate students at CMU. Guided them on their REU project, resulting in conference publication #23. The story of our collaboration was highlighted by the CMU news.
- Abigail Kamenetsky (2024-2025). Then a high school student, now double-majoring in Economics and Computer Science at University of Chicago. Guided her on implementing classical ML models in Python, and the design of a tree-search AI to play the PackIt! game featured in publication #11.
- Thomas Garrison (2024). Then an undergraduate student at CMU, now a PhD student in Mathematics at University of Florida. Co-authored conference publication #11.
- Aaron Gostein (2023-2024). Then an undergraduate student at CMU, now math content creator at the YouTube channel PurpleMind (46k+ subscribers) where I am featured in two long-form videos. Guided him on a research project in word combinatorics.
- Fernanda Gutiérrez (2023). Then an undergraduate student at University of Chile, now finishing a double-degree program at CentraleSupélec, France. Co-authored preprint #3.